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Artificial Intelligence and Machine Learning Question and Answer

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Long Question -1

As a Data Scientist at a digital platform like eSewa, unsupervised learning can provide significant value when dealing with millions of transactions and limited labeled data. Two high impact areas where it can deliver clear business value are:

- **Customer Segmentation**

A major business challenge is understanding different user behaviors for targeted marketing and customer retention. Without pre labeled user categories, the platform cannot distinguish between high value customers, inactive users, students, or merchants leading to generic campaigns that waste budget and reduce engagement.

To solve this, clustering techniques such as K-Means can be used. K-Means addresses this by grouping users based on features such as transaction frequency, average amount, merchant category diversity, and payment type. The elbow method can help determine the optimal number of clusters, producing segments like "power users," "inactive," or "frequent bill payers."

The model can run daily or weekly in a batch processing system using frameworks like Apache Spark. Generated cluster labels are stored in user databases and integrated with CRM systems to provide personalized offers, cashback rewards, and retention campaigns.

A major challenge is the need to predefine the number of clusters (k). If an incorrect value of k is selected, distinct user groups may be merged together, resulting in unclear or ineffective marketing segments. Additionally, K-Means assumes clusters are spherical in shape, which may not accurately represent real-world user behavior patterns. User behavior can also change over time due to seasonal trends or changing transaction habits, requiring periodic model retraining to maintain accuracy.

- **Fraud Detection**

Digital payment platforms are constant targets for evolving fraud schemes, such as account takeovers or wash trading. Fraud patterns evolve constantly and historical labeled fraud data remains scarce, making supervised approaches unreliable. So, it is important to identify transactions that are mathematically strange compared to the majority of legitimate traffic.

An effective approach is using Autoencoders for anomaly detection. Autoencoders offer an effective anomaly detection solution trained exclusively on normal transactions; the model learns to reconstruct legitimate behavior. And, when a fraudulent or highly unusual transaction occurs, the model fails to reconstruct it accurately, resulting in a high reconstruction error. For which features such as transaction amount, device fingerprint, location mismatch, and timing can be used as inputs.

This must be integrated into a Real-Time System. As a transaction is initiated, the model evaluates it in milliseconds. If the anomaly score exceeds a safe threshold, the system automatically triggers a step-up authentication like an OTP, biometric check or temporarily holds the funds for manual review.

A key challenge is the high false-positive rate, where unusual but legitimate transactions such as large medical payments may get incorrectly flagged, affecting customer experience.

Short Question

Overfitting Question 2

Overfitting occurs when a model learns training data too well, including noise and fails to generalize to new data. Which result in strong performance on training data but poor performance on unseen data. Two effective techniques to address this are dropout and early stopping.

- Dropout

Dropout randomly deactivates a percentage of neurons during each training step, forcing the network to avoid relying on any single neuron. This creates a more robust, distributed representation.

It acts as an ensemble method within a single model. At test time, all neurons are used, but their weights are scaled.

It is widely used in image classification to ensure CNN doesn't memorize specific pixel clusters.

- Early Stopping

Early Stopping observes the model's performance on a separate validation set. It monitors validation loss during training and halts the process once it stops improving, even if training loss continues to decrease.

This prevents the model from over adapting to training data beyond the point of useful learning.

It is used in Fraud detection, where patterns change rapidly, early stopping ensures the model generalizes to new transaction behaviors rather than just memorizing historical transaction records.

Neural Network Architecture Question 1

The difference between a Convolutional Neural Network (CNN) and a Recurrent Neural Network (RNN) is:

Feature	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Input Structure & Data Type	CNN process grid structured data where spatial relationships between pixels is the primary concern.	RNNs process sequential data such as text or time series where the order of inputs matters.
Information Flow & Design	Information flows in parallel through hierarchical layers of convolutions and pooling to detect local features simultaneously.	The architecture features recurrent loops where Information flows sequentially through a hidden state that carries context from earlier time steps into later ones.
Parameter Sharing Mechanism	Convolutional filters share parameters across spatial positions, detecting the same features regardless of where they appear.	Parameters are shared across time steps, meaning the same weights process each element in the sequence.

Real world application

- CNN: It is used for Image classification, Object detection, where spatial patterns like strokes, curves is important.
- RNN: It is used for Sentiment analysis, machine translation, where sequential context is essential.

Common Challenges

- Vanishing or exploding gradients:

During backpropagation, the gradients can either shrink to zero or grow uncontrollably, preventing the model from learning effectively.

This can be mitigated by using Batch Normalization to stabilize inputs or Gradient Clipping to cap the maximum size of a gradient during updates.

- Overfitting:

This occurs when a model becomes too complex and begins to memorize the specific noise and outliers of the training set rather than learning general patterns.

Techniques such as Early Stopping or dropout are used to halt the training process and randomly drop neurons.